MATH 202 Your name: _____

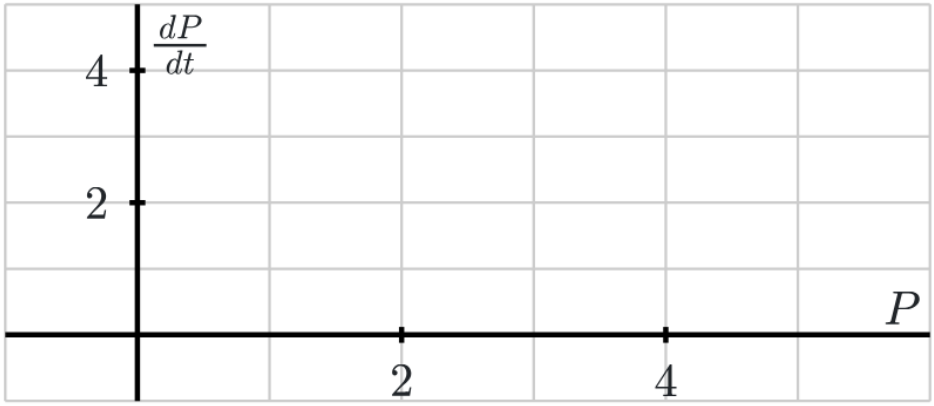
Learning target D2, version 1

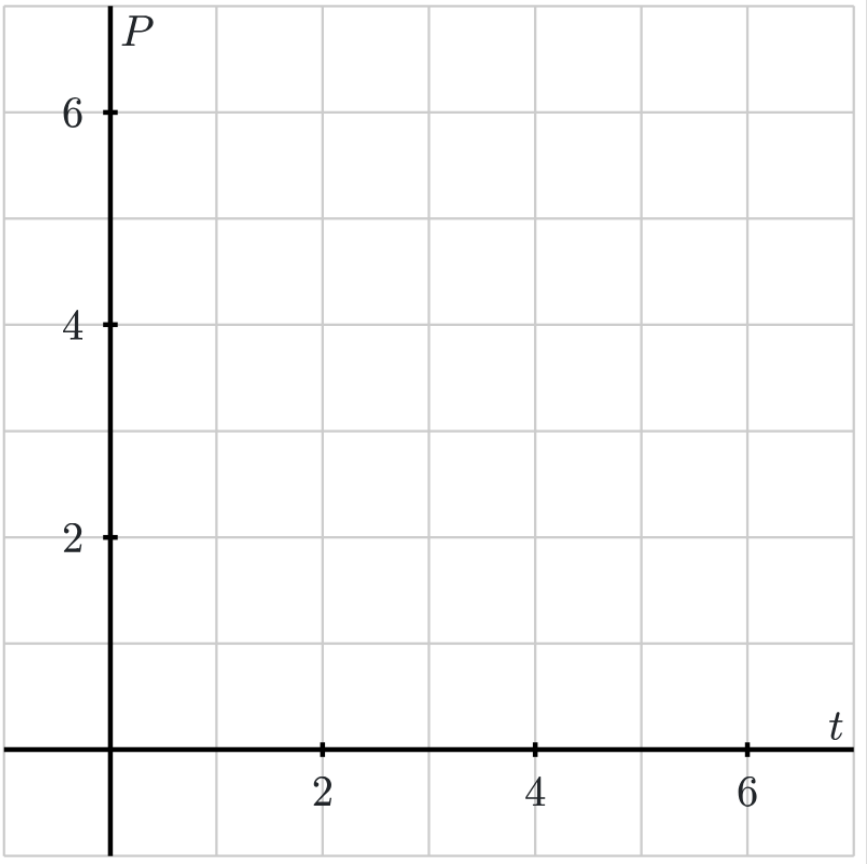
D2: Express real world situations as differential equations, determine any equilibrium solutions, and explain the significance of the equilibrium solutions in context.

A population of flamingos changes over time according to the differential equation

$$\frac{dP}{dt} = -\frac{1}{3}P(1-P)(5-P),$$

where P is measured in thousands (and t is probably measured in months or years).
Draw a phase graph and a slope field. Find all the equilibrium solutions, and explain how you know each one is stable or unstable.





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Learning target D3, version 1

D3: Apply Euler's Method to generate an approximate solution to an initial value problem, and explain the result's meaning in context.

A population of flamingos changes over time according to the differential equation

$$\frac{dP}{dt} = -\frac{1}{3}P(1-P)(5-P),$$

where P is measured in thousands (and t is probably measured in months or years).

(a) If $P(0) = 0.5$, use Euler's method with 3 steps to approximate $P(2)$. Record your work in a table. (Hint: If you're using 3 steps to get from $t = 0$ to $t = 2$, how big should Δt be?)

(b) Same thing, but this time, start with $P(0) = 1.5$.

(c) Compare your numerical results here with your graphical results on the D2 part of this problem. What recommendations do you have for flamingo keepers?